## Experiment 32: Bivariate Analysis and Linear Regression

**Aim:** To perform bivariate analysis and predict house prices using Linear Regression.

**Algorithm:**

1. Import required libraries.
2. Create house area and price data.
3. Calculate correlation.
4. Train a Linear Regression model.
5. Predict house price.
6. Evaluate the model and plot the data.

**Code:**

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

X = np.array([[1000], [1200], [1500], [1800], [2000]])
y = np.array([20, 24, 30, 36, 40])

print("Correlation:", np.corrcoef(X.flatten(), y)[0, 1])

model = LinearRegression()
model.fit(X, y)

pred = model.predict(X)

area = float(input("Enter house area: "))
print("Predicted Price:", model.predict([[area]])[0], "Lakhs")

print("MSE:", mean_squared_error(y, pred))
print("R2 Score:", r2_score(y, pred))
```

```
plt.scatter(X, y)
plt.plot(X, pred)
plt.xlabel("House Area")
plt.ylabel("Price")
plt.show()
```

**Input:**

Enter house area: 1600

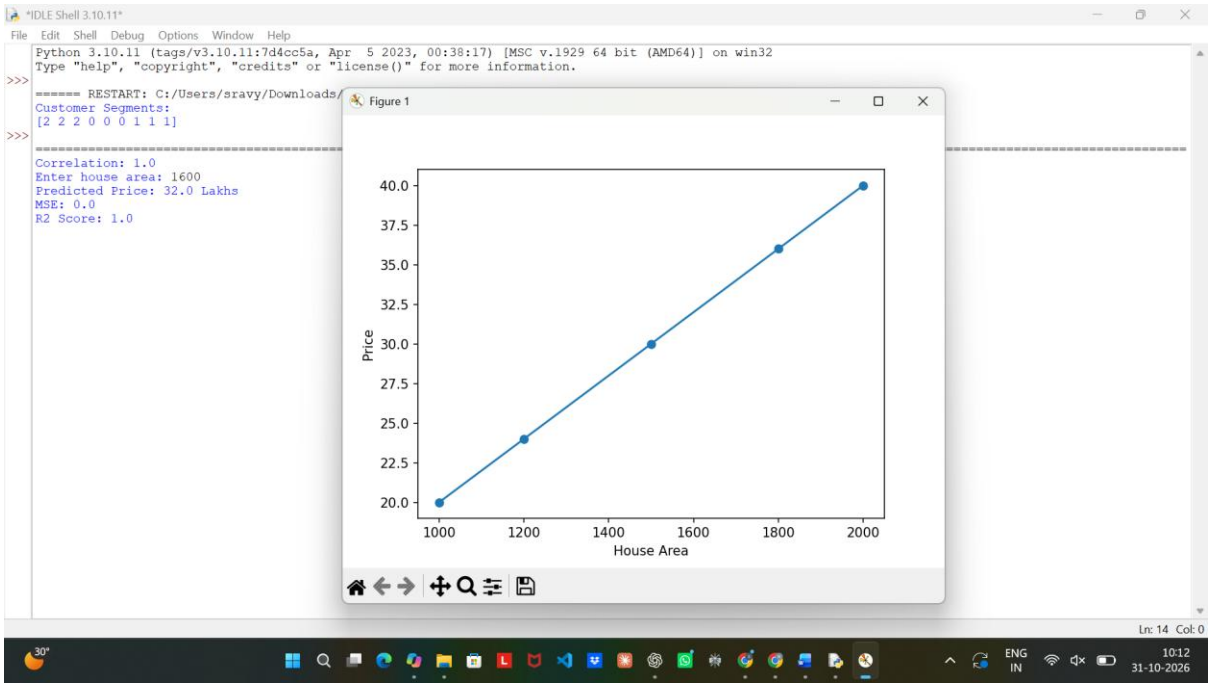
**Output:**

Correlation: 1.0

Predicted Price: 32.0 Lakhs

MSE: 0.0

R2 Score: 1.0




**Result:** Bivariate analysis and house price prediction were performed successfully.